

CAROLINA DEL PILAR DÍAZ

Senior AI Solutions Architect | Applied Scientist & Economist | Forward Deployed Engineer (FDE) Córdoba, Argentina | +54 9 351 554 2871 | caropilardiaz@gmail.com | [Linkedin](#)

PROFESSIONAL SUMMARY

Senior AI Architect & Applied Statistician with 20+ years of experience operating as a **strategic consultant (Internal & External)** for global enterprises. I bridge the gap between rigorous **econometric & statistical modeling** and **production-grade Agentic AI**. Unlike traditional developers, I architect and ship **end-to-end AI products (Backend & Frontend)** that are mathematically robust, **preventing model hallucinations** in regulated sectors. I specialize in **Multi-Agent Systems, Generative AI, and Causal Inference**, transforming complex theoretical frameworks into profitable, automated solutions. I have architected and sold **10+ full-stack AI applications**, combining the speed of **Low-Code (n8n, Antigravity)** with the rigor of advanced statistics, **Machine Learning, Deep Learning, and GenAI** across **Fintech, AgTech, Retail, Healthcare, Environment, Ecommerce, and Genetics**.

WORK EXPERIENCE

ITP PATAGONIA — Sr. AI Solutions Architect | Applied Statistician & Economist | Forward Deployed Engineer (FDE). 06/24 –Present | Remote. *Tech Stack: PyTorch, Transformers, HuggingFace, GPT/OpenAI API, LangChain, FastAPI, Langraph, Streamlit, Databricks, PySpark, AWS (SageMaker, Fraud Detector), Azure Framework Agent, Docker, n8n, AutoGen.* *Machine Learning, Deep Learning. Advanced Statistical Models. Bedrock, Strands, AWS agent AI.*

- **Multi-Agent Architecture & Cloud Orchestration (FDE Consultant):** Acting as a **Forward Deployed Engineer** for top-tier financial institutions (**Banco Galicia, ICBC**), I architected a proprietary **Multi-Agent Framework** and shipped **15+ autonomous AI apps**. Designed agnostic architectures that dynamically switch between **Google Gemini 1.5, GPT-4o, and Claude 3.5** via **n8n** on self-hosted servers, ensuring **data sovereignty** and reduced operational latency. ensuring **99.9%** output reliability and accelerating time-to-market by **~50%**.
- **Hybrid Fraud Detection System:** Engineered an anomaly detection and high-precision fraud engine combining **Generative AI** with **Traditional ML (LightGBM)**, **Isolation Forests**, and **Deep Learning (AENN)** on AWS/Azure. Reduced false positives and protected high-net-worth accounts by implementing **SHAP values** for model interpretability. preventing the erroneous lockout of **thousands of high-net-worth accounts monthly** and significantly improving client retention.
- **Predictive Analytics for Finance:** Developed predictive models for payroll clients and mutual fund investments using AWS SageMaker and AutoML, enabling efficient predictions for executive decision-making.
- **AI-Powered Retail Product Recommendation Engine:** I designed and deployed a next-generation recommendation system for retail products, **merging generative AI with advanced statistical probability** functions. This enabled hyper-personalization at massive scale, optimizing the match between the catalog and users, and increasing cross-selling rates and average order value by approximately 25%.
- **Scientific Validation in Production:** Solved critical failures in client AI initiatives where "black box" models were hallucinating. I inject rigorous **statistical validation layers** into the agent's reasoning loop to ensure output reliability for stakeholders.

NUBIRAL — Senior Data Scientist & Cloud AI Engineer (COE) (Oct 2023 – May 2024 | Remote). *Tech Stack: AWS Rekognition, AWS Lambda, S3, Azure OpenAI, LangChain, Azure Cognitive Search, Python, SQL, CI/CD.*

- **Computer Vision at Scale (Aerolíneas Argentinas):** Acting as a **Cloud Expert Commission member**, led the technical implementation of an object detection pipeline using **AWS Rekognition** and **GPT-4**. implemented integrations to automate maintenance inspections.

- **Enterprise GenAI Deployment (Banco Macro):** Designed and deployed secure RAG architectures utilizing **Azure Cognitive Search** and Form Recognizer to extract insights from massive unstructured PDF repositories without compromising banking data privacy.
- **Post-Call Analytics (Telecom S.A.):** Developed NLP pipelines (AWS Transcribe, Comprehend, QuickSight) to analyze thousands of daily customer interactions, calibrating sentiment models to improve customer service efficiency.

PI DATA STRATEGY & CONSULTING — Sr. Data Scientist, Statistician & Economist (Project Owner) (Mar 2021 – Oct 2023 | Córdoba). *Tech Stack: Azure ML, IBM Technologies, Python, R, Spark, Spatial Econometrics, NLP, Sequence Matcher, SQL. Advanced Statistical Models. Machine Learning. Deep Learning. Demography Analysis.*

- **Enterprise Data Consulting:** Acted as Lead Data Scientist and Project Owner for global Fortune 500 and government clients, including **Pernod Ricard, Anheuser-Busch InBev (Global & LATAM), Arcor S.A., GDM Seeds (Don Mario),** and the Argentine **Governments of Mendoza and Buenos Aires..**
- **Advanced NLP & Machine Learning:** Applied Deep Learning, NLP, and Sequence Matching algorithms using IBM and Azure technologies to create data-driven business solutions and text classification systems to improve management.
- **Georeferenced Demand Forecasting:** I implemented a hybrid architecture based on **Autoencoders for user segmentation (Deep Clustering) and LightGBM/XGBoost for geographically** accurate sales forecasting. This solution allowed us to identify unmet demand niches through the app, improving forecast accuracy by 30% and increasing the logistics NPS for key corporate clients..

PREVIOUS SCIENTIFIC AND ACADEMIC LEADERSHIP (2012 – 2021 | Córdoba)

- **INTA (Head of Statistical Research / Principal Statistician):** Pioneered deep learning (CNN) and computer vision applied to drone spectral imagery in AgTech. Automated crop health monitoring, reducing data analysis time by approximately 50%. Applied Generalized Linear Mixed Models (GLMM) to massive genomic datasets, validating experimental designs across all departments and ensuring the absence of errors in published and peer-reviewed data.
- **National University of Córdoba (Professor and Principal Investigator):** Taught deep learning and computer vision courses in the Data Science program. Mentored students in predictive modeling and advanced spatial statistics.

EDUCATION

- **Diploma in Data Science, Machine Learning & Applications** | UNC.(2020)
- **Master of Science in Applied Statistics (c)** | Universidad Nacional de Córdoba (2008)
- **Licenciatura (Bachelor's) in Economics** | Universidad Nacional de Córdoba (1998)

CERTIFICATIONS & MEMBERSHIPS

- **Certifications:** Agentic AI on AWS (BeSA Candidate, 2026) | Generative AI Engineer Associate (Candidate, 2026) | Azure AI Fundamentals (Candidate, 2026) | AWS Academy Cloud Architecting (2025) | Generative AI with LLMs (DeepLearning.AI, 2023) | MLOps Fundamentals CI/CD (2022) | Databricks & Apache Spark (2022).
- **Memberships:** Gen AI Community (2025-Present) | AWS Girls Argentina (2025-Present) | Mujeres en Tecnología (MET) (2019-Present) | International Biometric Society (IBS) (2006-Present).

SELECTED PUBLICATIONS & TALK

- **Speaker:** "Agentic AI in Practice: Bridging the Gap Advanced Statistical models and LLMs" - LinkedIn Tech Talks (2025).
- **Conference Speaker:** Regular oral exhibitor at the International Biometric Society (IBS) Annual Meetings (2016-2024), Generative AI and Advanced Statistical Models Course.2025.

